Sportsturf Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Sportsturf Science (A08603)

Short Title:	Sportsturf		
Faculty	Science and Computing		
Department	Science		
Cluster	Horticulture		
Author	GMCMAHON		
Credits	10	**Level:**	Intermediate

Description of Module / Aims

This module aims to enable students to develop an understanding of the construction and maintenance of turf and artificial surfaces used for sports.

Programmes

SPOR-0092	BSc in Horticulture (WD_SHORT_D)	2 / 3 / E
SPOR-0092	BSc in Horticulture (WD_SHORT_D)	3 / 5 / E
SPOR-0092	BSc in Horticulture (WD_SHORT_DX)	2 / 3 / E
SPOR-0092	BSc in Horticulture (WD_SHORT_DX)	3 / 5 / E

Indicative Content

- Turfgrass science: Soils for sportsturf, drainage and irrigation for sportsturf, grass morphology and physiology, turfgrass pathology and entomology, sportsturf weeds and control
- Turfgrass maintenance: Turfgrass mowing, management of organic matter, nutrition of turfgrass, aeration and compaction relief of turfgrass, renovation and repair of turfgrass, topdressing and materials, growth regulation, and other maintenance techniques
- Construction and maintenance of a range of sportsturf facilities: Sports pitches, bowling greens, cricket grounds, lawn tennis courts, equestrian sports and artificial surfaces

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine and specify appropriate soil types and root zones for specific sportsturf construction.
2. Interpret the principles of drainage and irrigation and describe appropriate methods for sportsturf.
3. Distinguish the main turf grass species and a range of pests, diseases and weeds associated with turfgrass.
4. Explore the factors influencing pests of turfgrass and propose appropriate IPM control methods in response to turfgrass pests, diseases and weeds.
5. Explore maintenance techniques and equipment required for turfgrass maintenance.
6. Demonstrate a range of turfgrass maintenance techniques.
7. Produce a plan of appropriate maintenance techniques for a range of sportsturf constructions.

Learning and Teaching Methods

- Lectures
- Practical assignments
- Reports

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4,5,
Continuous Assessment	50%	
Assignment	25%	7
Practical	25%	3,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

40%-49%: Demonstrates a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Logically and competently carries out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrates sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrates authoritative handling of complex material and a highly developed and extensive understanding.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	72	
Practical	24	
Independent Learning	174	

Supplementary Material

- Brown, S. _Sports Turf and Amenity Grassland Management_. Wiltshire: The Crowood Press Ltd, 2005.
- Evans, R.D.C. _Bowling greens: Their history, construction and maintenance_. Bingley, West Yorkshire, England: Sports Turf Research Institute, 1992.

- Evans, R.D.C. _Cricket grounds: The evolution, maintenance construction of natural turf cricket tables and outfields_. Bingley, West Yorkshire, England: The Sports Turf Research Institute, 1991.
- Evans, R.D.C. _Winter games pitches, the construction and maintenance of nat__ural turf pitches_. 1st. Bingley, West Yorkshire, England: Sports Turf Research Institute, 1994.
- Fry, J. and B. Huang. _Applied Turfgrass Science and Physiology_. New Jersey: Wiley, 2004.
- Lodge, R. and S. Shanks. _All Weather Surfaces for Horses_. 3rd ed. London: J. A. Allen Publishing, 2005.
- McCarty, L.B., I.R. Rodriguez, B.R. Bunnell and F.C. Waltz. _Fundamentals of Turfgrass and Agricultural Chemistry_. New Jersey: Wiley, 2003.
- Perris, J. _Grass Tennis Courts. How to construct and maintain them_. 1st ed. Bingley, West Yorkshire, England: The Sports Turf Research Institute, 2000.
- Phulla, J., J. Krans and M. Goatley. _Sports Fields: A manual for design, construction and maintenance_. 2nd ed. New Jersey: Wiley, 2010.

Requested Resources

- Room Type: Biology Lab